SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 6 **Total Marks:30**

Annexure A

Scenario-Based

Code of Conduct:

I _E. Sakthi priya (192520041)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Sakthi priya

Annexure A

Scenario-Based

1. Use Case Diagram Scenario

A **Smart Pharmacy Management System** enables customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines, while administrators manage medicine inventory and user accounts. The interactions among system users are not clearly defined.

Based on this scenario, analyze the system requirements, identify all actors and use cases, and design a **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to ensure complete system functionality.

(5 Marks)

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2. Class Diagram Scenario

A **Smart Hotel Management System** consists of entities such as Guest, Room, Reservation, Payment, and Receptionist. The existing design does not clearly define class attributes, operations, or relationships among these entities.

Based on this scenario, identify the required classes, define suitable attributes and methods, establish relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a detailed **Class Diagram**.

(5 Marks)

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3. Sequence Diagram Scenario

In an **Online Banking System**, a customer logs in, transfers funds, verifies the transaction using OTP, and receives a transaction confirmation. The communication among the customer, authentication service, banking server, and database is incomplete.

Based on this scenario, analyze the sequence of operations, identify participating objects and message flows, and design a **Sequence Diagram** representing the complete interaction process.

(5 Marks)

4. Activity Diagram Scenario

A **Passport Application System** allows applicants to submit applications, upload supporting documents, complete identity verification, pay processing fees, and receive passports. The workflow does not clearly represent approval, rejection, or concurrent activities.

Based on this scenario, analyze the workflow, identify decision points and parallel activities, and design an optimized **Activity Diagram** using decision nodes, forks, and joins.

(5 Marks)

5. State Diagram Scenario

A **Smart Elevator Control System** operates through states such as Idle, Door Open, Door Closed, Moving Up, Moving Down, Emergency Stop, and Maintenance Mode. The transitions between these states are not clearly specified.

Based on this scenario, identify all possible states, triggering events, and transitions, and construct a **State Transition Diagram** that accurately represents the elevator's behavior.

(5 Marks)

6. Deployment Diagram Scenario

A **Smart Home Automation System** uses smart sensors, IoT controllers, a home gateway, cloud servers, and a mobile application to monitor and control lighting, security, and appliances remotely. The deployment architecture does not clearly illustrate hardware nodes, deployed software, or communication channels.

Based on this scenario, analyze the deployment environment, identify all hardware nodes and software artifacts, and design a **Deployment Diagram** showing the physical architecture and communication links.

(5 Marks)

RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Scenario: Smart Pharmacy Management System

Actors

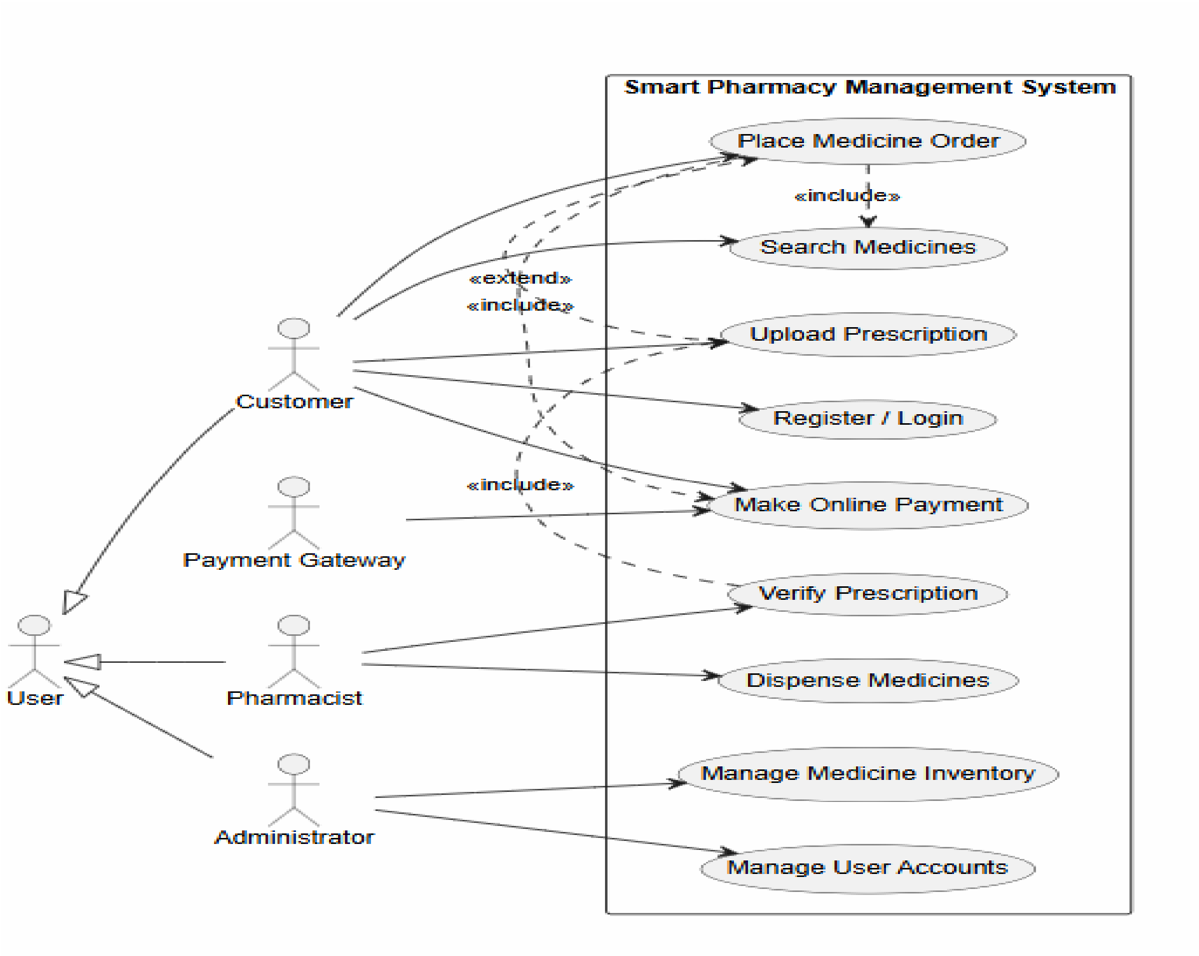
1. Customer
2. Pharmacist
3. Administrator
4. Payment Gateway (External System)

Use Cases

- Register/Login
- Search Medicines
- Upload Prescription
- Place Medicine Order
- Make Online Payment
- Verify Prescription
- Dispense Medicines
- Manage Medicine Inventory
- Manage User Accounts

Relationships

- **<<include>>**
  - Place Medicine Order → Search Medicines
  - Place Medicine Order → Make Online Payment
  - Verify Prescription → Upload Prescription
- **<<extend>>**
  - Upload Prescription → Place Medicine Order (only when prescription medicines are ordered)
- **Generalization**
  - Customer, Pharmacist, and Administrator are specialized users of the **User** actor.



<u>2. Scenario: Smart Hotel Management System</u>

Classes

1. Guest
2. Room
3. Reservation
4. Payment
5. Receptionist
6. Person (Parent Class)

<u>Attributes and Methods</u>

Person

Attributes

- personId
- name
- phone

Methods

- login()
- logout()

Guest

Attributes

- guestId
- address

Methods

- bookRoom()
- cancelReservation()

<u>Receptionist</u>

Attributes

- employeeId
- shift

Methods

- checkIn()
- checkOut()
- assignRoom()

<u>Room</u>

Attributes

- roomNo
- roomType
- price
- status

Methods

- checkAvailability()
- updateStatus()

<u>Reservation</u>

Attributes

- reservationId
- checkInDate

- checkOutDate

Methods

- createReservation()
- cancelReservation()

<u>Payment</u>

Attributes

- paymentId
- amount
- paymentMethod

Methods

- makePayment()
- generateReceipt()

<u>Relationships</u>

Inheritance

- Person → Guest
- Person → Receptionist

Association

- Guest books Reservation.
- Receptionist manages Reservation.

Aggregation

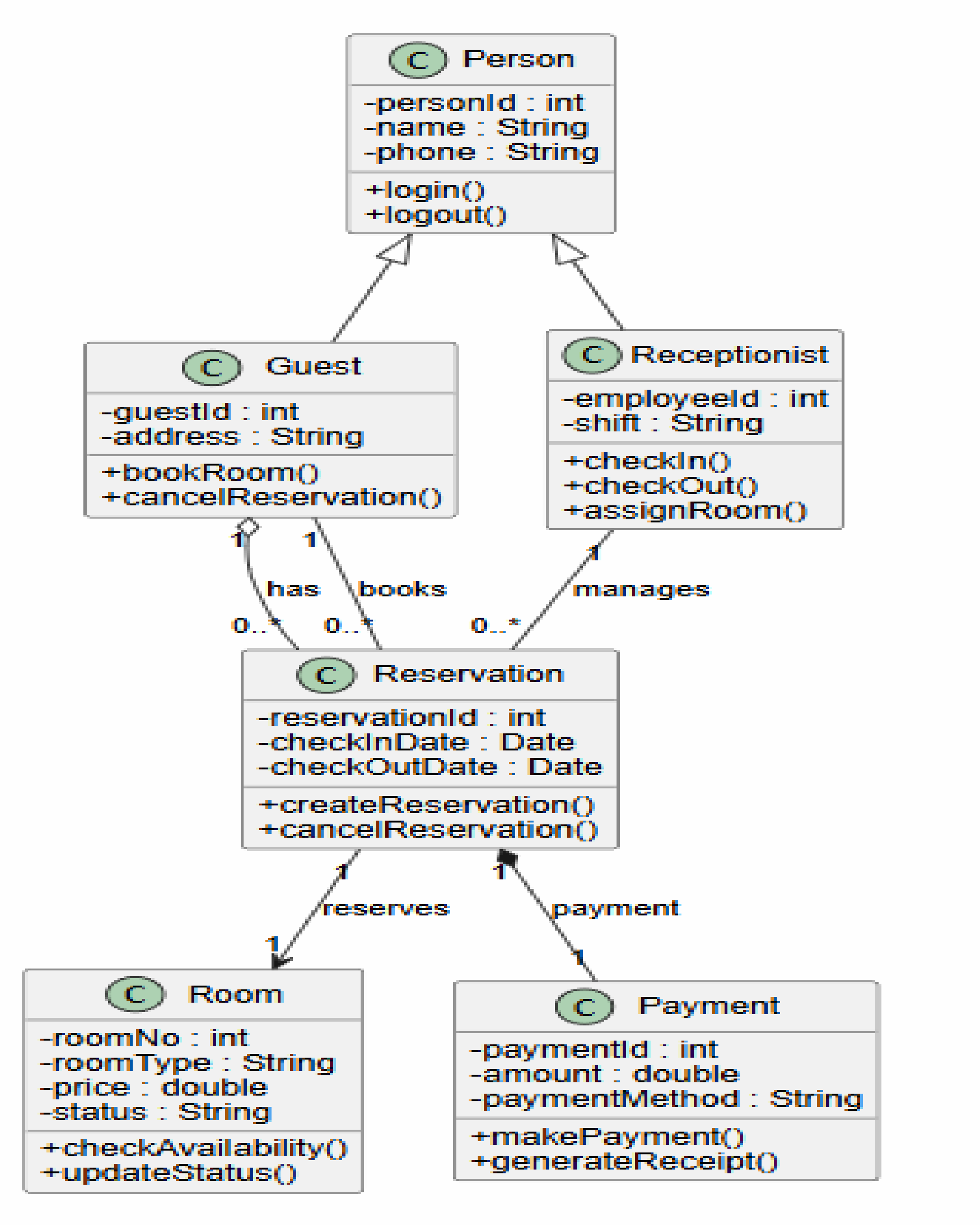
- Guest has Reservations.

Composition

- Reservation contains Payment.

Multiplicity

- One Guest → Many Reservations (**1..***)
- One Reservation → One Room (**1**)
- One Reservation → One Payment (**1**)
- One Receptionist → Many Reservations (**1..***)



3. Scenario: Online Banking System

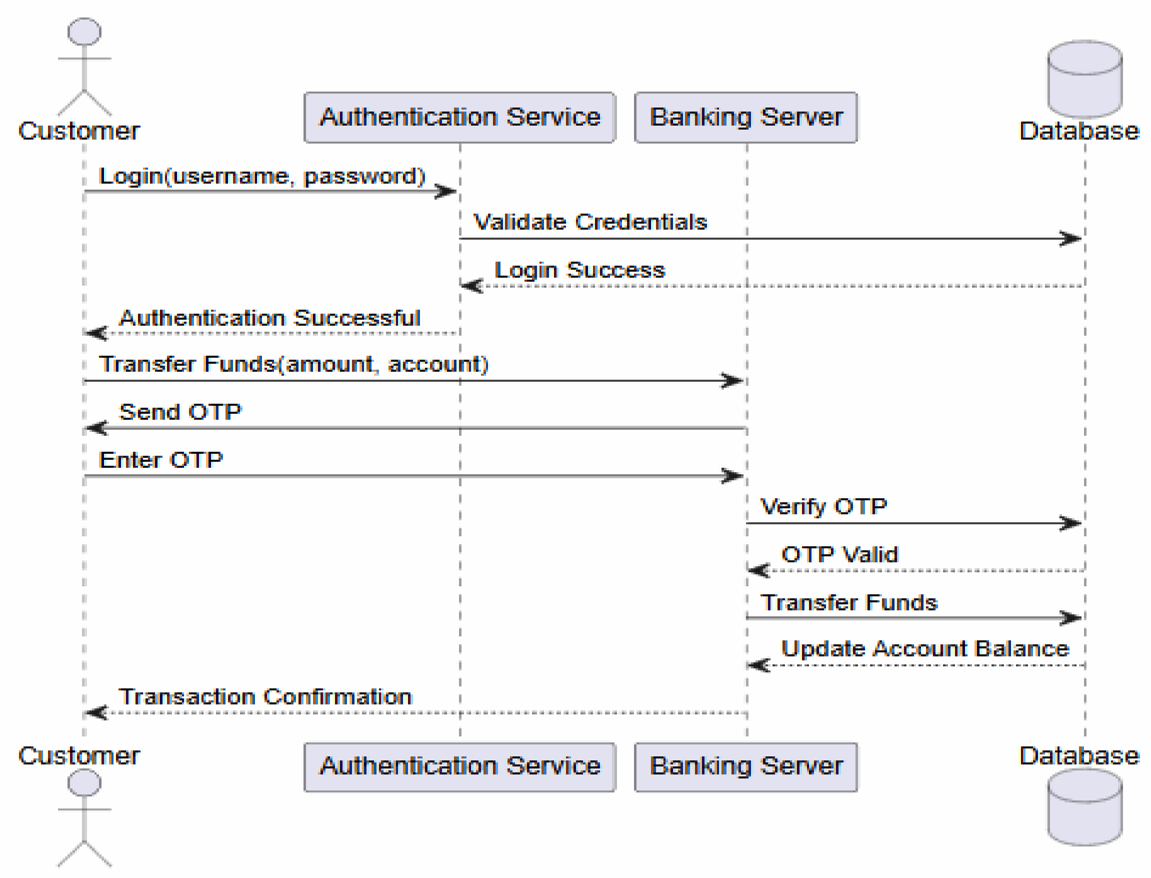
Participating Objects

1. Customer
2. Authentication Service
3. Banking Server
4. Database

Sequence of Operations

1. Customer enters login credentials.
2. Authentication Service validates the login details.

3. Authentication Service checks the Database.
4. Database returns the authentication result.
5. Customer requests fund transfer.
6. Banking Server generates and sends an OTP.
7. Customer enters the OTP.
8. Banking Server verifies the OTP with the Database.
9. Database confirms the OTP.
10. Banking Server transfers the funds.
11. Database updates the account balance.
12. Banking Server sends a transaction confirmation to the Customer.



4. Scenario: Passport Application System

Workflow

1. Applicant submits the passport application.
2. Applicant uploads the required documents.

3. The system performs **Identity Verification** and **Fee Payment** simultaneously (parallel activities).
4. After both activities are completed, the application is reviewed.
5. A decision is made:
    - **Approved** → Passport is issued.
    - **Rejected** → Applicant is notified of rejection.
6. The process ends.

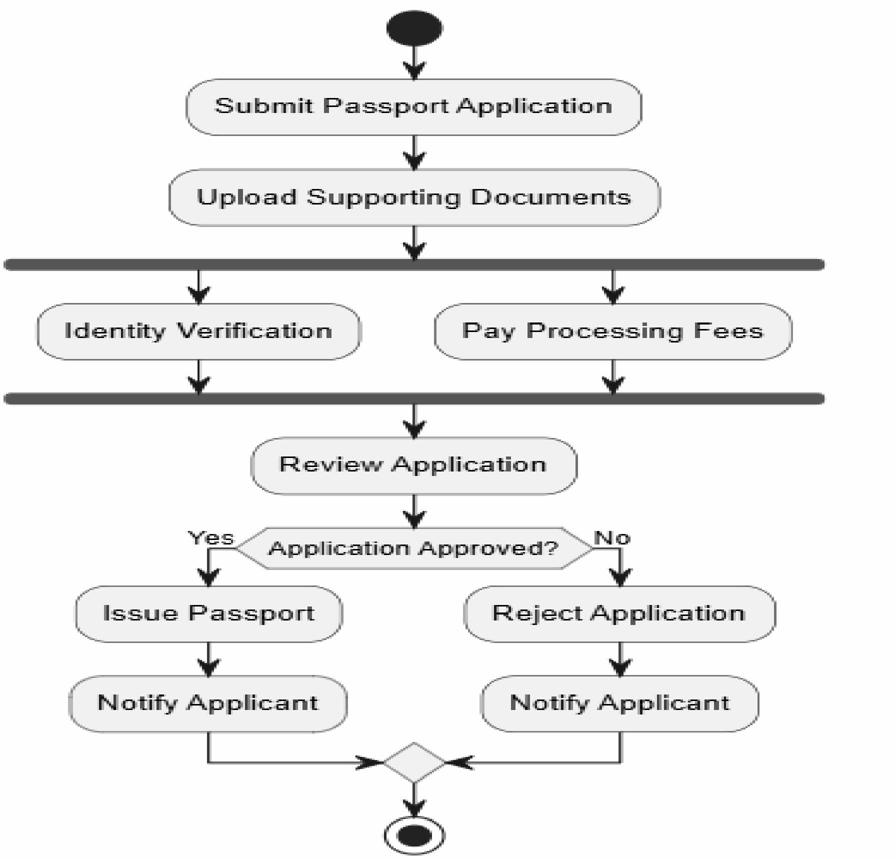
Decision Points

- Application Approved?
    - **Yes** → Issue Passport.
    - **No** → Reject Application.

Parallel Activities

- Identity Verification
- Payment of Processing Fees

These activities are executed at the same time using **Fork** and **Join** nodes.



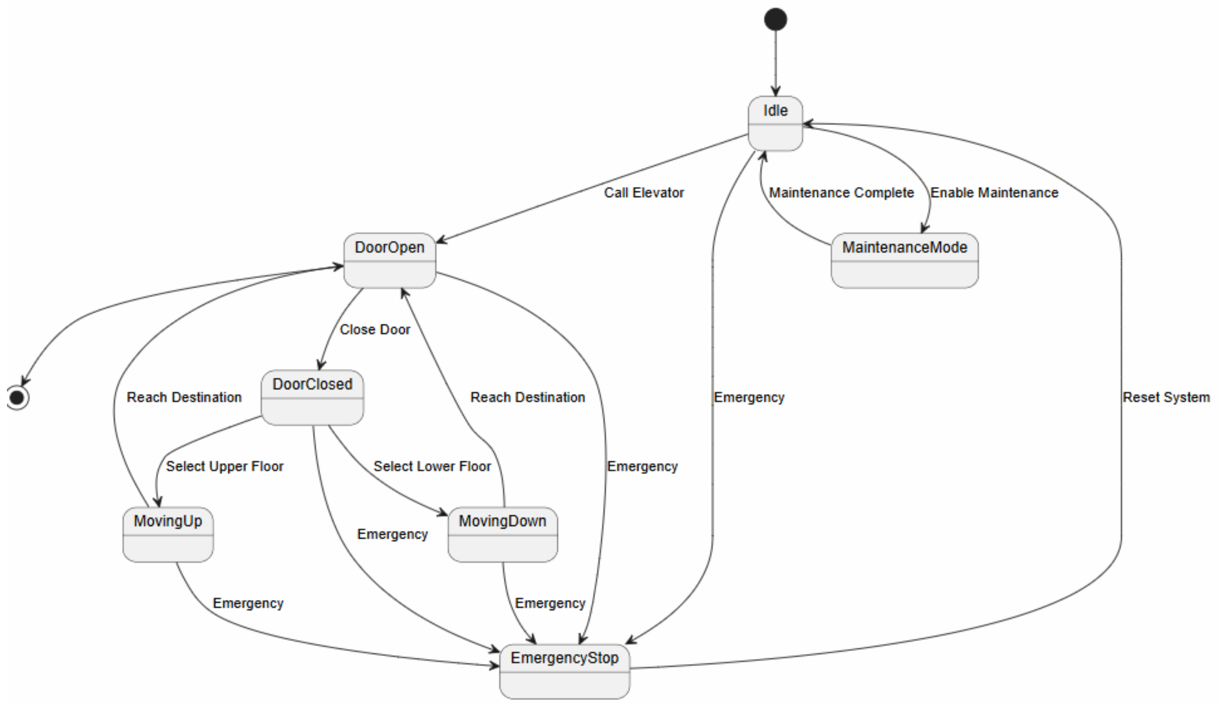
<u>**5. Scenario: Smart Elevator Control System**</u>

States

1. Idle
2. Door Open
3. Door Closed
4. Moving Up
5. Moving Down
6. Emergency Stop
7. Maintenance Mode

Triggering Events

- **Call Elevator** → Idle to Door Open
- **Close Door** → Door Open to Door Closed
- **Select Upper Floor** → Door Closed to Moving Up
- **Select Lower Floor** → Door Closed to Moving Down
- **Reach Destination** → Moving Up/Moving Down to Door Open
- **Emergency Button Pressed** → Any state to Emergency Stop
- **Reset System** → Emergency Stop to Idle
- **Maintenance Enabled** → Idle to Maintenance Mode
- **Maintenance Completed** → Maintenance Mode to Idle



6. Scenario: Smart Home Automation System

Hardware Nodes

1. Smart Sensors
2. IoT Controller
3. Home Gateway
4. Cloud Server
5. Mobile Device

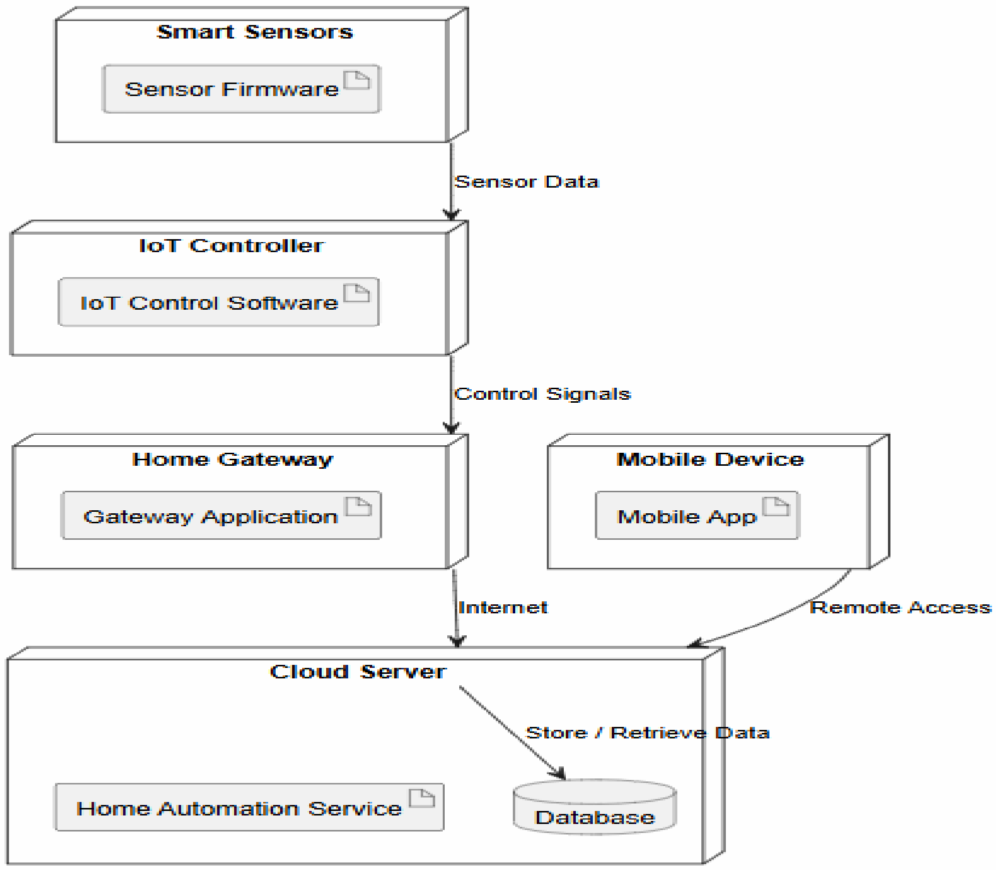
Software Artifacts

- Sensor Firmware
- IoT Control Software
- Gateway Application
- Home Automation Service
- Database
- Mobile App

Communication Links

- Smart Sensors ↔ IoT Controller
- IoT Controller ↔ Home Gateway
- Home Gateway ↔ Cloud Server (Internet)
- Mobile App ↔ Cloud Server
- Cloud Server ↔ Database

The deployment diagram shows how software components are deployed on hardware devices and how they communicate to monitor and control home appliances remotely.


Smart Sensors
Sensor Firmware
Sensor Data
IoT Controller
IoT Control Software
Control Signals
Home Gateway
Gateway Application
Mobile Device
Mobile App
Internet
Remote Access
Cloud Server
Store / Retrieve Data
Home Automation Service
Database